

Prathvi Raj M V

Phone: 7349343800 | Email: prathvirajmv@gmail.com | LinkedIn | GitHub

EDUCATION

JSS Science and Technology University

Bachelor of Engineering – Electronics and Communication Engineering; CGPA: 9.35

Mysore, India

2023 – 2027

EXPERIENCE

Western Digital – Firmware Summer Intern

Bangalore, India

NAND Subsystem Validation and Testing

- Executed firmware validation for NAND storage subsystems by analyzing SSD metadata and verifying firmware functionality across multiple test scenarios.
- Developed Python-based parsing and visualization tools for NAND metadata analysis, reducing manual debugging effort and improving validation workflows.
- Investigated SSD data structures, firmware logs, and storage architecture while contributing two Invention Disclosure Forms (IDFs) for future HDD improvements.

PROJECTS

6 DOF Robotic Arm

C++, Python, MATLAB, Fusion 360

- Designed and fabricated a 6-DOF robotic manipulator using steppers, magnetic encoders, and custom mechanical assembly.
- Implemented forward/inverse kinematics, trajectory planning, and embedded firmware enabling synchronized multi-axis motion with accurate end-effector positioning.

Rocket Flight Computer

STM32, C++, KiCad, Fusion 360

- Designed and developed an STM32-based flight computer implementing Thrust Vector Control (TVC) and Active Fin Stabilization for high-power rocketry. Implemented IMU sensor fusion, PID-based attitude stabilization, telemetry communication, onboard data logging, and real-time control firmware for autonomous flight.

Delta Robotic Arm

Python, Fusion 360

- Engineered a high-speed 3-DOF Delta robot for pick-and-place automation using lightweight parallel kinematics. Developed forward/inverse kinematics, workspace analysis, and motion simulation to improve positioning accuracy and trajectory execution.

2-Axis CNC Plotter

STM32, GRBL, Embedded C, Fusion 360

- Designed and fabricated a GRBL-based CNC plotting system with custom mechanical assembly and motion control.
- Developed a CAD-to-G-code workflow enabling accurate vector drawing through synchronized stepper motor control.

SKILLS

Programming: C, C++, Python, Embedded C, MATLAB, SQL

Embedded Systems: STM32, ESP32, Arduino, Raspberry Pi, ARM Cortex-M, GPIO, UART, SPI, I²C, PWM, ADC, Timers, Interrupts, Sensor Interfacing

Firmware & Storage: Firmware Validation, NAND Flash, SSD Architecture, Storage Validation, Embedded Testing

Robotics & Control: Forward/Inverse Kinematics, PID Control, Sensor Fusion, IMU Integration, Motion Planning, Trajectory Generation, Robotics, Computer Vision

Artificial Intelligence: Machine Learning, Deep Learning, CNNs, LLMs, PyTorch, Scikit-learn, Pandas,

Software Development: Git, Jira, GitHub, Linux, Flask, HTML, CSS

CAD / Simulation: Fusion 360, KiCad, MATLAB, Simulink

HACKATHONS & ACHIEVEMENTS

- **Winner – Creat-a-thon 7.0 (IEEE PES, DSCE, Bengaluru):** Developed an autonomous railway track inspection robot using computer vision and embedded sensing for defect detection and track monitoring.
- **Winner – TECHNOTSAV (VVCE, Mysuru):** Built an IoT-enabled autonomous floating platform for real-time industrial wastewater quality monitoring with wireless telemetry.
- **Winner – EPOCH '26 (VIT, Bengaluru):** Designed a multi-sensor IoT monitoring system with machine learning-based anomaly detection for smart water distribution networks.
- **Winner – Project Expo (DSC-JSSSTU, Mysuru):** Designed and demonstrated an STM32-based Active Fin Stabilization system with real-time attitude control for high-power rocketry.
- **BIRAC E-YUVA Fellowship '26 (Anna University):** Selected for developing the *Smart Telemetric Hydro-Storage Orthodontic Appliance*, an IoT-enabled medical device for intelligent wet/dry appliance storage and monitoring.

PATENTS / PUBLICATIONS

- **Bio-Inspired Fin Design for Controlled Steering in Rocketry** (Patent) –Under Review, App. No. 202541059467.
- **Smart Telemetric Hydro-Storage Orthodontic Appliance Case** (Patent) –Under Review, App. No. 202641063439.
- **Hybrid-Adhesion Six-Legged Swarm Robot for Spacecraft and Habitat Inspection** – Publication: IEEE Space.

CERTIFICATIONS

- Python, C++, Machine Learning, Fusion 360 (Udemy) • Java, Git (LinkedIn Learning) • Deep Learning (NPTEL, IIT Ropar, 80/100)